

Natural Language Processing for Ancient Greek

Design, advantages and challenges of language models

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Computational methods have produced meaningful and usable results to study word semantics, including semantic change. These methods, belonging to the field of Natural Language Processing, have recently been applied to ancient languages; in particular, language modelling has been applied to Ancient Greek, the language on which we focus. In this contribution we explain how vector representations can be computed from word co-occurrences in a corpus and can be used to locate words in a semantic space, and what kind of semantic information can be extracted from language models. We compare three different kinds of language models that can be used to study Ancient Greek semantics: a count-based model, a word embedding model and a syntactic embedding model; and we show examples of how the quality of their representations can be assessed. We highlight the advantages and potential of these methods, especially for the study of semantic change, together with their limitations.

Keywords: Ancient Greek, semantic change, computational linguistics, language models, Natural Language Processing, word embeddings, semantic space

1. Introduction: Language modelling for Ancient Greek

Linguistic research on corpora of ancient texts has recently seen a growing interest in the use and development of quantitative and computational methods, broadly falling into the field of Natural Language Processing (NLP). NLP encompasses the techniques allowing for automated, computer-based quantitative

analysis of linguistic data, often relying on machine learning. NLP methods allow for a large-scale analysis of language data which, paired with an expert view, can enhance our understanding of linguistic and cultural phenomena to a degree that manual inspection alone cannot achieve (McGillivray 2014: 211). Indeed, NLP techniques make it possible to extract information about language usage in corpora and identify possibly meaningful patterns by performing highly complex calculations efficiently. One aspect where they have been shown to be particularly helpful is semantic analysis (Lenci & Sahlgren 2023: 15–19), especially through language modelling. The focus of this contribution is on how to exploit the potential of language modelling for studying semantic aspects of Ancient Greek, including semantic change, as well as highlighting its limitations.

Language models are computed representations based on large quantities of texts (“training data”). During the training phase of a model, information about word sequences in the training data is obtained. The resulting model can then predict which word is the most likely to follow a given word. The most recent language models are extremely large and sophisticated, and they can perform complex tasks such as text summarisation, question answering and translation. In smaller language models, such as those presented here, a numerical representation (“vector”) is generated for each word based on its contexts of occurrence in a given corpus during model training, the phase in which the model learns patterns from the training corpus. In line with the distributional hypothesis (Harris 1954), it is expected that words occurring in similar contexts will exhibit semantically similar vectors. Two words that occur in similar contexts may be synonyms, but also antonyms, or more broadly semantically related, for example “shoe” and “sock”. Word vectors can be used, via simple comparison calculations, to find out which words have a similar distribution. Although occurrences in similar contexts can be due to a variety of reasons, vector-based analysis has focused mostly on lexical semantics, for example to compare the meaning or usage of different words or to detect and track semantic change (some examples are Gulordava & Baroni 2011; Kulkarni et al. 2015; Hamilton et al. 2016; Kozłowski et al. 2019).

Since NLP methods work on a statistical basis, the amount of available linguistic data and their quality is crucial. In particular, ensuring that the corpus used to obtain the word vectors is as representative of the target language variety as possible is important. If the aim is to perform semantic change analysis, the corpus must cover different time periods. Despite the currently available corpora for ancient languages being smaller than the corpora of modern languages (which can include billions of words), recent research on Ancient Greek (e.g., Boschetti 2009; Rodda et al. 2017; Perrone et al. 2019, 2021), Latin (Sprugnoli et al. 2020) and Sanskrit (Sandhan et al. 2023) has shown that language models can also yield usable results for ancient languages.

In this study we explain and illustrate ways of employing language models in the study of Ancient Greek semantics, and we discuss the advantages and limitations of these approaches. In §2, we introduce corpus annotation and present some annotated corpora of Ancient Greek, while in §3 we introduce distributional spaces built from word co-occurrences and we give examples of information that can be extracted from them. In §4, we compare count-based and word embedding models. In §5, we present a selection of computational studies on semantic change in Ancient Greek. In §6, we explain the importance of evaluating the performance of language models, and we show an example of how two different language models trained on Ancient Greek (a count-based and a word embedding model) can be evaluated against a benchmark. Finally, in §7, we introduce and evaluate a syntactic model trained on Ancient Greek, and we compare its results to those obtained with the other two tested models.

2. Annotation and existing annotated corpora of Ancient Greek

“Annotation” is the operation of associating each token in a corpus to a piece of linguistic information, e.g., lemma, morphological analysis, or part of speech. Lemmatisation, the process that associates each inflected form in a corpus to its base form in a dictionary (“lemma”), is often unnecessary when working with modern languages, due to the large amount of corpus data available. However, lemmatisation is a common practice to alleviate data sparsity when training a language model on ancient languages, which tend to be highly inflected and to have a limited number of available texts. Compared to a non-lemmatised corpus, a lemmatised corpus of Ancient Greek contains a lower number of forms (“types”) for the same total number of instances (“tokens”). Types thus have higher frequencies of occurrence, allowing for more reliable statistics. Of course, a language model trained only on lemmas ignores morphology, making it impossible to extract information about the behaviour of inflected forms¹ and possibly introducing undesired ambiguities.

Existing annotated corpora of Ancient Greek can be directly processed by language models. For example, the Diorisis Ancient Greek Corpus (Vatri & McGillivray 2018) and the Lemmatized Ancient Greek Texts released by Giuseppe Celano² are annotated with lemmas, parts of speech and morphological analysis, while the PROIEL (Haug & Jøhndal 2008) and the Ancient Greek Dependency

1. Note that inflected forms of the same lemma can have very different, even non-overlapping, distributions (see for example Tognini-Bonelli 2001: 92).

2. <https://github.com/gcelano/LemmatizedAncientGreekXML>

Treebank (Bamman & Crane 2011) include syntactic dependency information (both are “treebanks”, i.e., corpora of texts with syntactic annotation). Even if the most common approach with ancient languages is to train models on lemmatised corpora, it is indeed possible to include more annotation information during the training, as we will see later with syntactic embeddings, including syntactic annotation in the training data.

3. Distributional spaces

The simplest way to obtain word vectors is to count the number of times that a word co-occurs with every other word in a corpus. For example, for a sample corpus composed only of the sentence “To be, or not to be, that is the question”, co-occurrence counts are stored in the following co-occurrence matrix, where the numbers in each row count how many times a “context” word in a column occurs immediately to the right or to the left of a word in a row (“target” word):

Table 1. Co-occurrence matrix for a corpus formed by the sentence “To be or not to be, that is the question”

	Context words							
	to	be	or	not	that	is	the	question
to	0	2	0	1	0	0	0	0
be	2	0	1	0	1	0	0	0
or	0	1	0	1	0	0	0	0
not	1	0	1	0	0	0	0	0
that	0	1	0	0	0	1	0	0
is	0	0	0	0	1	0	1	0
the	0	0	0	0	0	1	0	1
question	0	0	0	0	0	0	1	0

Every string of numbers in a row is a vector, a numerical condensation of the co-occurrences of its target word. For example, the vector of *be* is (2,0,1,0,1,0,0,0). The length of the “context window” is one of the parameters that can be set when training a language model. A window of one was chosen in the example above for simplicity, but in real cases a larger window is usually adopted, for example five words to the left and five to the right of each word. Vectors can be used as coordinates to represent words as points in a geometric space with *n* dimensions (axes), where *n* represents the number of digits of the vector. In the case of the

matrix above, words can be represented in an eight-dimensional space (reflecting the number of unique words in the matrix).³ The space is also called “semantic space” because words are located according to their semantics, captured by their vectors. Words occurring in similar contexts are therefore closer in space. A key feature of semantic spaces is that semantic relatedness between words (i.e., their belonging to the same semantic field) can be quantified by measuring the angle between the lines connecting two word-points to the origin of the axes. Indeed, a word vector can also be represented as an arrow starting from the origin of the axes and pointing at the specific word-point in the geometric space. It is thus possible to rank the words in a corpus according to their level of semantic relatedness to a chosen target word; an often adopted metric is the cosine similarity.

For example, the ten nearest neighbours to the target word *ναῦς* (*naus*) ‘ship’, extracted together with their cosine similarity to *ναῦς* from a language model⁴ trained on Ancient Greek texts,⁵ are: *πλέω* (*pleō*) ‘sail’ (0.37), *τριήρης* (*triērēs*) ‘trireme’ (0.37), *ναός* (*naos*) ‘ship’⁶ (0.35), *πλοῖον* (*ploion*) ‘floating vessel’ (0.31), *ἐπιπλέω* (*epipleō*) ‘sail upon’ (0.28), *λιμὴν* (*limēn*) ‘harbour’ (0.28), *ναυτικός* (*nau-tikos*) ‘naval’ (0.28), *παραπλέω* (*parapleō*) ‘sail by’ (0.28), *στόλος* (*stolos*) ‘(sea-)voyage, seaforce’ (0.27), *ναυμαχία* (*naumachia*) ‘sea-fight’ (0.27). These words all clearly belong to the semantic field of navigation. Another example is the ten nearest neighbours to the target word *κακός* (*kakos*) ‘bad’, all belonging to a semantic area of badness, unhealthiness and moral evaluation, with the exception of *ἄνθρωπος* (*anthrōpos*) ‘man’: *πάσχω* (*paschō*) ‘suffer, feel’ (0.23), *ἀγαθός* (*agathos*) ‘good’ (0.22), *αἰσχρός* (*aischros*) ‘shameful’ (0.20), *νόσος* (*nosos*) ‘sickness’ (0.19), *ἄνθρωπος* (*anthrōpos*) ‘man’ (0.18), *φίλος* (*philos*) ‘loved’ (0.18), *συμφορά* (*symphora*) ‘misfortune’ (0.18), *ἀπόλλυμι* (*apollymi*) ‘destroy, kill’ (0.18), *δεινός* (*deinos*) ‘terrible, marvellous’ (0.18) and *ἄδικος* (*adikos*) ‘unjust’ (0.18).

3. For more details and a graphic representation, see McGillivray (2022: 5–6).

4. Count-based model weighted with Positive Pointwise Mutual Information, with a context window of five words. Stopwords were filtered out from the corpus (stopwords are high-frequency words which generally have a more grammatical than lexical meaning). The model was trained with software from the LSCDetection repository at <https://github.com/Garrafao/LSCDetection> (Kaiser et al. 2021). The models trained for this work (not all discussed in this article) can be found at <https://doi.org/10.5281/zenodo.8369516>.

5. The Diorisis Ancient Greek Corpus (Vatri & McGillivray 2018).

6. This is actually a genitive form of *ναῦς*, erroneously lemmatised in the Perseus Digital Library, hence in the Diorisis corpus, as if it was a nominative.

4. Count-based models and word embeddings: Potential and limitations

Counting co-occurrences is the easiest and most interpretable way to obtain word vectors. However, the vectors obtained have the disadvantage of being long (i.e., having as many digits as the word types in the corpus) and sparse (i.e., with many zeroes, corresponding to all the cases in which a target word never co-occurs with a certain context word). Non-frequency-based techniques to compute word vectors also exist, for example word embeddings, which leverage artificial neural networks. The basis for the computation is always word co-occurrences, but instead of just counting them, the model stores more complex information about probabilities of co-occurrence. An example of word embedding technique is Word2vec (Mikolov et al. 2013), available in two versions, called Continuous Bag-of-Words and Continuous Skip-gram. Compared to count-based models, word embeddings produce denser and shorter vectors, which are easier to handle and yield high-quality results. The models learn more complex semantic relationships than the simple relatedness between pairs of words. They store semantic knowledge about proportions of the kind *woman : queen = man : king*. Thus, given any three words in the analogy, they return the final fourth. For example, a word embedding model⁷ trained on a corpus of Classical Ancient Greek texts⁸ fills the “x” slot in the proportion [ὅσιος (*hosios*) ‘pious’ : δίκαιος (*dikaios*) ‘just’ = ἀνόσιος (*anosios*) ‘impious’ : x] with ἄδικος (*adikos*) ‘unjust’.

Although word embeddings allow the exploration of deeper semantic relationships, to produce reliable representations they require larger training corpora than count-based models. For this reason, count-based models have usually been preferred in most research in language modelling for Ancient Greek. The need for a larger quantity of training data is thus a limitation to the application of word embeddings to ancient languages. More generally, due to the underlying statistical approach to language, a limitation of all language models is that they do not produce usable results for low-frequency words in the corpus, as observed by Boschetti (2009: 62) for words occurring less than four times in the training data.

7. A Word2vec Continuous Bag-of-Words model trained according to the CADE framework (Bianchi et al. 2020; repository available at <https://github.com/vinid/cade>) with a context window of five words.

8. Including all the texts from the Diorisis Corpus that were written between 499 and 324 BCE.

5. Computational studies on semantic change in Ancient Greek

Language models have also been employed for the detection and tracking of lexical semantic change by comparing vector representations for the same word in different time periods (see e.g., Kulkarni et al. 2015; Hamilton et al. 2016; Di Carlo et al. 2019). To be able to perform automatic semantic change detection, a diachronic corpus must be available and large enough to train language models on different temporal subcorpora of it. For Ancient Greek, the viability of performing automatic semantic change has recently been confirmed. The first studies applying (count-based) language models to study the semantics of Ancient Greek used the Thesaurus Linguae Graece (TLG) as a training corpus (Boschetti 2009; Rodda et al. 2017).

Boschetti (2009) built⁹ two semantic spaces on two diachronic slices of the TLG (representing the BCE and CE portions) and tested their usability for the study of semantic change. It is indeed possible to compare not only vectors of different words in one semantic space, but also vectors of the same word in different spaces; in this case in two spaces built on the BCE and CE subcorpora. Different vector representations for the same target word in two (or more) diachronic corpora indicate that the target co-occurs with different words in different time periods, indicating a possible change in its semantics. Boschetti chose as case studies *θάλασσα* (*thalassa*) ‘sea’ and *θάνατος* (*thanatos*) ‘death’, hypothesising that the concrete term *θάλασσα* (*thalassa*) ‘sea’ would not have undergone semantic change between the BCE and CE periods, but that an abstract term such as *θάνατος* (*thanatos*) ‘death’ might have as a result of a changing cultural and religious context. An analysis of the words that are the most strictly related in meaning to *θάλασσα* (*thalassa*) ‘sea’ and *θάνατος* (*thanatos*) ‘death’ (i.e., the words that are the nearest neighbours in the semantic space) confirmed this hypothesis and the viability of the method: many of the nearest neighbours to *θάλασσα* (*thalassa*) were shared in both the BCE and CE periods (they were almost all geographic terms), while for *θάνατος* (*thanatos*), the nearest neighbours changed significantly, and in the CE era they contained terms with “a strong Christian valence” (Boschetti 2009: 63–64).

Rodda et al. (2017) created the first system for automatic lexical semantic change detection for Ancient Greek. Like Boschetti (2009), they divided the corpus into two portions, corresponding to the BCE and CE eras, respectively. Then, they trained a language model on each subcorpus, thus building two semantic spaces which they used to analyse the semantic change of words. Great differences between the BCE and CE vectors for the same word meant that the

9. By using the Infomap software package, available at <https://infomap-nlp.sourceforge.net/>.

word was used in different contexts in the two periods, which may indicate a change in the word's semantics. The authors then studied the words for which the semantic shift was the most significant, as indicated by the largest differences between the BCE and CE vectors for the same words. Some words that changed significantly acquired a Christian meaning or connotation, such as *παραβολή* (*parabolē*), changing from 'comparison' to 'parable', *λαός* (*laos*), changing from 'people' to 'the Christians', and *κύριος* (*kyrios*), changing from 'lord, master' to 'Lord, God'. More unexpectedly, an example of a technical term with a significant semantic shift is *σύμπτωμα* (*sympptoma*). An inspection of the nearest neighbours for each subcorpus clarifies that the use of this word shifted from a predominantly philosophical context to the medical domain, having originally meant 'property' and subsequently shifting to mean 'symptom' (Rodda et al. 2017: 18). The words whose meanings changed the most across the two subcorpora were also graphically visualised in a geometric space¹⁰ (Rodda et al. 2017: 22). These graphic visualisations allowed the researchers to individuate clusters of words that underwent greater semantic change, corresponding to very specific semantic fields, such as medicine, astronomy, geometry and moral philosophy. Although semantic spaces proved to be useful for the study of semantic change, a critical interpretation of results is needed since a change in the contexts of occurrence of a word through time does not necessarily mean that semantic change has occurred, but could be just due to the different sampling of the diachronic subcorpora.

Perrone et al. (2019) addressed the issue of semantic change by building GASC (Genre-Aware Semantic Change), a topic model for the detection of semantic change in Ancient Greek. GASC uses the distributional information of words in the form of word co-occurrence data and couples it with information about the genre of the texts. The topic model then generates a given number of "topics" (to be interpreted as word senses) as clusters of similar words. Genres play an important role in the semantics of words, particularly in ancient languages. For example, the meanings of the polysemous word *μῦς*, (*mys*) 'mouse', 'mussel' or 'muscle' vary not only over time, but also depending on which text types they are used in, particularly in technical (specifically medical) and non-technical texts. Perrone et al. (2021) compare the performance of GASC with that of a genre-independent topic model and of models using word embeddings. GASC tends to outperform the other models, which shows the importance of accounting for textual features such as genre in the computational study of semantic change.

10. The authors used the t-distributed Stochastic Neighbor Embedding (t-SNE) technique, which makes it possible to represent multidimensional vectors in two dimensions.

The studies just discussed focused on testing the applicability of NLP methods to Ancient Greek; therefore, most of them did not attempt to increase our knowledge of the semantics of those words since they were chosen as easily interpretable case studies to assess the reliability of the language models. Future studies building on these works should aim to expand our knowledge of the semantics of Ancient Greek words with new observations.

6. Evaluation of the performance of language models

When dealing with semantic change, a critical approach towards the results of language models is not the only necessary component. More generally, it is imperative to evaluate the quality of the knowledge acquired by the models before we can draw any conclusions from them. Language models can indeed be built with different architectures and parameters for their training: for example, the context window can be tuned differently. Evaluations of the quality of the vector representations produced by language models are usually measured against “benchmarks”: resources expressly created (normally by hand) for evaluating model performance. They contain examples of “correct answers” that we could expect the model to output: for example, *κακός* (*kakos*) ‘bad’ could be included in a benchmark as a good example of nearest neighbour for *αἰσχρός* (*aischros*) ‘shameful’. A publicly accessible benchmark allows for the evaluation of different models against the same “gold standard”.

Rodda et al. (2019) carried out the first attempt to evaluate count-based models of Ancient Greek against already existing resources: an ancient lexicon, a modern lexicon, and the Ancient Greek WordNet (Boschetti et al. 2016).¹¹ Table 2 gives an example of evaluation of two different language models trained on the Diorisis Corpus (a count-based model and a Word2vec model) against items from Ancient Greek Relatedness Embeddings Evaluation (AGREE), a benchmark for the evaluation of language models of Ancient Greek.¹² More precisely, it collects gold standard judgements about word relatedness. The metric used for the evaluation, “precision”, is a way to quantify the overlap between nearest neighbours extracted from a language model and a gold standard:

11. For a discussion of the results and limitations of such an approach, see Stopponi et al. (2024).

12. <https://doi.org/10.5281/zenodo.8027490>. The benchmark was created from expert judgements and expert evaluations of model outputs, but the gold standard used for this contribution consists of only expert judgements.

$$precision = \frac{n.\text{nearest neighbours output by the model which are also in the benchmark}}{n.\text{nearest neighbours extracted}}$$

The nearest neighbours overlapping with the gold standard in the benchmark are given in bold:

Table 2. Evaluation of the performance of a count-based model weighted with Positive Pointwise Mutual Information and a Word2vec Continuous Bag-of-Words model on two lexical items against the AGREE benchmark. Details of the training are provided respectively in §3 and §4.

Lemma	Model	Precision	Five nearest neighbours from models	Related lemmas from the AGREE benchmark
ἐλεύθερος (eleutheros) ‘free’	Count-based PPMI	0.8	δοῦλος (doulos) ‘slave’; ἐλευθερία (eleutheria) ‘freedom’; δουλεύω (douleuō) ‘to be a slave’; πολίτης (politēs) ‘citizen’; δεσπότης (despotēs) ‘master’	δοῦλος (doulos) ‘slave’; πολίτης (politēs) ‘citizen’; ἐλευθερία (eleutheria) ‘freedom’; δούλιος (doulios) ‘slavish’; παρρησιαστικός (parrēsiastikos) ‘outspoken’; πολιτικός (politikos) ‘related to citizens’; ἐκών (ekōn) ‘wittingly’; ἴσος (isos) ‘equal’; Ἑλλάς (Hellas) ‘Greece’; δουλεύω (douleuō) ‘to be a slave’; ἐκκλησία (ekklēsia) ‘assembly’; ἀδούλευτος (adouleutos) ‘one who has never been a slave’; αὐτόνομος (autonomos) ‘independent’; ἄφρετος (aphetos) ‘let loose, free (from work)’; ἀνέπαφος (anepaphos) ‘untouched, unharmed’; δέσμιος (desmios) ‘binding, bound’; ἄδεια (adeia) ‘freedom from fear’; δημοκρατία (dēmokratia) ‘democracy’; Ἑλλήν (Hellēn) ‘Greek’; ἄνθρωπος (anthrōpos) ‘man’; πολιτεία (politeia) ‘citizenship’

Table 2. (continued)

Lemma	Model	Precision	Five nearest neighbours from models	Related lemmas from the AGREE benchmark
σῆμα (<i>sēma</i>) 'sign, mark'	Word2vec	0.2	δοῦλος (<i>doulos</i>) 'slave';	
	CBOW		πένης (<i>penēs</i>) 'poor'; δεσπότης (<i>despotēs</i>) 'master'; ἄτιμος (<i>atimos</i>) 'deprived of civic rights'; πτωχός (<i>ptōchos</i>) 'beggar'	
	Count-based	0	τεύχω (<i>teuchō</i>) 'make, build'; γαῖα (<i>gaia</i>) 'land'; μιν (<i>min</i>) 'him'; πέλω (<i>pelō</i>) 'come into existence, become'; ὄφρα (<i>ophra</i>) 'that, in order that, while, until'.	τάφος (<i>taphos</i>) 'funeral, tomb'; σῶμα (<i>sōma</i>) 'body'; σύμβολον (<i>symbolon</i>) 'token, guarantee'; σημαίνω (<i>sēmainō</i>) 'indicate'; τεκμήριον (<i>tekmērion</i>) 'proof'; μνημεῖον (<i>mnēmeion</i>) 'memorial'; εἶδωλον (<i>eidōlon</i>) 'phantom, image'; τύμβος (<i>tymbos</i>) 'tomb'; γράμμα (<i>gramma</i>) 'letter'; χέω (<i>cheō</i>) 'pour'; λήκυθος (<i>lēkythos</i>) 'oil-flask'; γνώρισμα (<i>gnōrisma</i>) 'mark, token' φάσμα (<i>phasma</i>) 'apparition, phantom'; σύνθημα (<i>synthēma</i>) 'signal, token, sign'; μνήμα (<i>mnēma</i>) 'memorial'; σημαντικός (<i>sēmantikos</i>) 'significant'; θαύμα (<i>thauma</i>) 'wonder'; δηλώω (<i>dēlōō</i>) 'to show'; θεός (<i>theos</i>) 'god'; θάνατος (<i>thanatos</i>) 'death'; στήλη (<i>stēlē</i>) 'block of stone'
	Word2vec	0	νέκυσ (<i>nekys</i>) 'corpse'; ἄλωή (<i>alōē</i>) 'threshing floor, prepared garden'; σπέος (<i>speos</i>) 'cavern'; πτολίεθρον (<i>ptoliethron</i>) 'citadel'; οὔδας (<i>oudas</i>) 'ground'	

The example of ἐλεύθερος (*eleutheros*) 'free' shows how language models with different architectures and parameter settings can yield different results, even if they are trained on the same corpus. Moreover, comparing the results for ἐλεύθερος (*eleutheros*) 'free' and σῆμα (*sēma*) 'sign, mark' shows how the capacity of a language model to produce good semantic representation can vary over the lexicon. Word frequency surely influences the quality of the semantic repre-

sentations, but other factors could play a role, for example, the level of polysemy of a word or the concreteness of its meaning. Furthermore, another downside of NLP methods is that the results extracted from language models are not always explicable, i.e., it is not always possible to understand why a model produces a certain output. Indeed, Lenci and Sahlgren (2023: 219–220) describe word vectors produced by neural networks as “black box representations”.

7. Syntactic word embeddings

As stated above, information other than lemmas or word forms can be taken into account in the training of language models. Since at least Levy and Goldberg (2014), it has been noticed that incorporating syntactic information in the training of word embeddings can change the type of linguistic relations captured by the models, compared to simple lemma- or token-based word vectors. By taking into account the dependency structure of sentences, syntactic embeddings should capture more functional and less topical (or domain) similarity. Using an example from Levy and Goldberg (2014), given the word *Florida*, a count-based model is more likely to return counties and cities in Florida as nearest neighbours (i.e., meronyms, in a relationship of topical similarity with the input word), whereas dependency-based embeddings may return other US states (i.e., co-hyponyms, in a relationship of functional similarity with *Florida*). Moreover, syntactic embeddings can be expected to capture similarity between words sharing a syntactic function and/or part of speech: for example, nouns often used as subjects are more likely to have as nearest neighbours not only words that occur in similar contexts, but in particular other nouns often used as subjects.

Experiments on Ancient Greek vector space modelling have yet to exploit the relative wealth of syntactically annotated data available. To show the difference in results between syntactic word embeddings, count-based- and word-embedding models, we present an example from some preliminary results we obtained by modifying an existing model architecture incorporating syntactic information during the training of word embeddings (following the methodology in Al-Ghezi and Kurimo 2020). The method starts from trees which encode the syntactic information contained in a treebank in parenthetical form, each corresponding to one sentence in the treebank, and converts them into a network graph. The graphs are then merged into a SuperGraph, consisting of the union of all graph-converted parenthetical trees. Consider, for example, the sentences in

(1a) and (2a) from our Ancient Greek corpus,¹³ their labelled parenthetical trees in (1b) and (2b) and the same parenthetical trees after lemmatisation and stopword removal (see footnote 4) in (1c) and (2c).¹⁴

- (1) a. καὶ μεταυτικά συνέμισγε τὰ στρατόπεδα πεζῇ καὶ
and immediately.after meet.3SG.IMPF the.NOM army.NOM by.land and
νηυσί
ship.PL.DAT
b. (καὶ ((μεταυτικά)_{ADV} ((συνέμισγε)_{VP} (τὰ (στρατόπεδα))_{NP} ((πεζῇ) καὶ
(νηυσί)_{ADV})))
c. (μεταυτικά (συμμείγνυμι (στρατόπεδον) (πεζῇ ναῦς)))
immediately.after meet army by.land ship
'And immediately afterwards the armies met in close fight, on land and
sea.'
(Herodotus, *Histories*, 5.112)
- (2) a. πεζῇ δέ, ὥς συνῆλθε τὰ στρατόπεδα,
by.land PTC when gather.3SG.AOR the.NOM army.NOM
συμπεσόντα ἐμάχοντο
charge.AOR.PTCP.NOM.PL fight.3PL.IMPF
b. ((πεζῇ)_{ADV} (δέ)_{PTC} (ὥς (συνῆλθε)_{VP} (τὰ (στρατόπεδα))_{NP})_{ADV}
(συμπεσόντα)_{XADV} (ἐμάχοντο)_{VP}))
c. (πεζῇ ((συνέρχομαι (στρατόπεδον))) συμπίπτω μάχομαι)
by.land gather army charge fight
'On land, when the armies met, they charged and fought.'
(Herodotus, *Histories*, 5.112)

The SuperGraph generated from the parenthetical trees in (1c) and (2c) is shown in Figure 1.

13. The corpus of Ancient Greek used to carry out this experiment includes the following treebanks: the PROIEL treebank (Haug & Jøhndal 2008), the Gorman treebank (Gorman 2020), the PapyGreek treebank (Vierros & Henriksson 2021), the Pedalion treebank (Keersmaekers et al. 2019) and the Ancient Greek Dependency Treebank (Bamman & Crane 2011). The code used to merge them after normalising their different annotation schemes, pre-process the corpus and train the graph-based models is openly available at <https://github.com/npedrazzini/ancientgreek-syntactic-embeddings>.

14. The parentheses divide each sentence into a hierarchy corresponding to the syntactic relation between the constituents. The format of the parenthetical trees in (1c) and (2c) is typical "Tree" objects in the Python package Natural Language Toolkit (NLTK) and is not meant to be easily close-read, but only to serve as input for their conversion into graphs. Also note that the labels ADV (adverbial phrase), XADV (adverbial phrase sharing an argument with the main clause), VP (verbal phrase) and NP (noun phrase) in (1b) and (2b) are only meant to make the relations between the constituents more transparent without reflecting any particular syntactic framework.

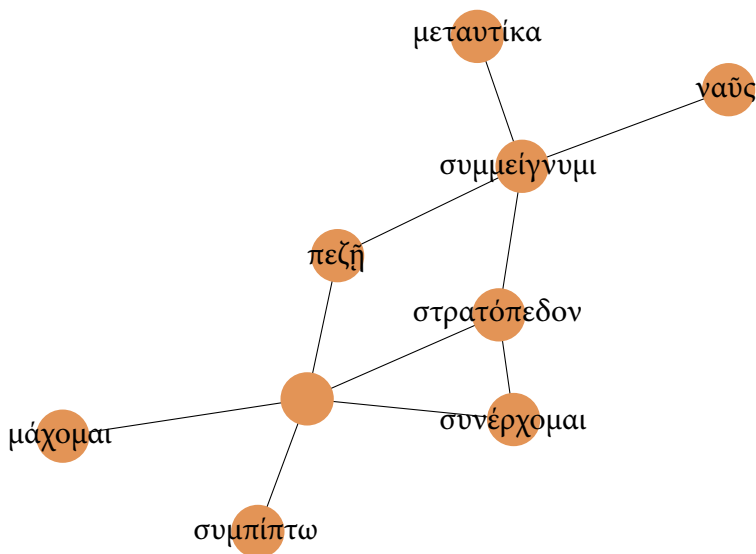


Figure 1. Example of a SuperGraph from the parenthetical trees in (1c) and (2c)

The SuperGraph for the whole corpus is then used as input for the Node2vec model (Grover & Leskovec 2016),¹⁵ a modification of the Word2vec Skip-gram architecture which trains word vectors for each node in the graph. An example of the different kinds of semantic relations captured by embeddings which incorporate syntactic information, compared to traditional models, is provided in Table 3, showing the five nearest neighbours of ἐλεύθερος (*eleutheros*) ‘free’ according to a graph-based model and to the same count-based- and word embedding models used in the previous sections.

Most nearest neighbours returned by the count-based and the word embeddings model belong to the semantic field of “free” and “freedom” or to their topical scope.¹⁶ On the other hand, although the nearest neighbours returned by the graph-based model can also be considered to be broadly within the semantic field of “free”, we see that their relationship with ἐλεύθερος (*eleutheros*) is more functional than topical. They are all adjectives, which can be expected to show similar distributions as ἐλεύθερος (*eleutheros*). The example just seen, showing the dif-

15. As a preliminary experiment, we trained our graph-based model by keeping the default parameters proposed by the authors by setting the window size to five and not filtering out any low-frequency words.

16. The nearest neighbours belong to different parts-of-speech: for example δουλεύω (*douleuō*) ‘to be subject’ is a verb, δεσπότης (*despotēs*) ‘master’ is a noun and ἄτιμος (*atimos*) ‘deprived of civic rights’ is an adjective. Of course, there are also lemmas that can be used both as nouns and as adjectives, such as δούλος (*doulos*) ‘slave’.

Table 3. Comparison between the five nearest neighbours of ἐλεύθερος (*eleutheros*) ‘free’ from a count-based-, a word embeddings- and the graph-based model

Count-based model	Word embeddings model (Word2vec CBOW)	Graph-based model
δοῦλος (<i>doulos</i>) ‘slave’; ἐλευθερία (<i>eleutheria</i>) ‘freedom’; δουλεύω (<i>douleuō</i>) ‘to be subject’; πολίτης (<i>politēs</i>) ‘citizen’; δεσπότης (<i>despotēs</i>) ‘master’	δοῦλος (<i>doulos</i>) ‘slave’; πένης (<i>penēs</i>) ‘poor’; δεσπότης (<i>despotēs</i>) ‘master’; ἄτιμος (<i>atimos</i>) ‘deprived of civic rights’; πτωχός (<i>ptōchos</i>) ‘beggar’	αἰσχρὸς (<i>aischros</i>) ‘shameful’; λωίων (<i>loiōn</i>) ‘better’; εὐσχήμων (<i>euschēmōn</i>) ‘graceful’; εὐτελής (<i>eutelēs</i>) ‘worthless’; πλούσιος (<i>plousios</i>) ‘wealthy’

ferent types of neighbours returned by different model architectures for the same target word, suggests that the quality of semantic representations produced by different architectures should be evaluated according to different criteria, and *ad-hoc* benchmarks are probably needed for the specific types of models. When we compare syntactic word embeddings with classic embeddings or count-based models, for example, we cannot simply talk about differences in quality of vector representations since they capture different semantic relations to start with. Using precision calculated against a benchmark (as seen in §6) as a proxy for the quality of a model is thus a useful approximation only if one is privy to how the benchmark was built and how the models to evaluate are going to be used.

As a practical example of how using precision on the graph-based model to evaluate its quality against the same benchmark used above (see Table 2) results in a very low score, the five nearest neighbours of the word Ἀθηναῖος (*Athēnaios*) ‘Athenian’ extracted from our graph-based model can be compared with the semantically-related lemmas to the same word from the AGREE benchmark (Table 4).

Among the nearest neighbours from the graph-based model, only one is also found in the benchmark, Λακεδαιμόνιος (*Lakedaimonios*) ‘Lacedaemonian’, which would give us a low precision of $1/5 = 0.2$. However, the syntactic model itself does perform well: it simply captures functional rather than topical similarity and returns only proper names and demonyms, which, of course, could be considered as very useful results, depending on the task at hand.

8. Conclusions

In this contribution we have presented and compared some computational language models that can be used to study the semantics of Ancient Greek words,

Table 4. The five nearest neighbours of Ἀθηναῖος (*Athēnaios*) ‘Athenian’ from our graph-based model compared to the semantically-related lemmas for the same word from the benchmark

5-nearest neighbours of Ἀθηναῖος (<i>Athenaios</i>)	Precision	Lemmas from the AGREE benchmark
Λακεδαιμόνιος (<i>Lakedaimonios</i>) ‘Lacedaemonian’; Πλαταιεύς (<i>Plataieus</i>) ‘Plataean’; Φασάηλος (<i>Phasaēlos</i>) ‘Phasael’; Ἀθήναιος (<i>Athēnaios</i>) ‘Athenaeus’; Κλεομένης (<i>Kleomenēs</i>) ‘Cleomenes’	0.2	πόλις (<i>polis</i>) ‘city’; ἀνὴρ (<i>anēr</i>) ‘man’; πολιτεία (<i>politeia</i>) ‘citizenship’; Ἀθῆναι (<i>Athēnai</i>) ‘Athens’; Ἀττικὸς (<i>Attikos</i>) ‘Attic’; Λακεδαιμόνιος (<i>Lakedaimonios</i>) ‘Lacedaemonian’; πολυπράγμων (<i>polypragmōn</i>) ‘busy about many things’; δικαστής (<i>dikastēs</i>) ‘judge’; συκοφάντης (<i>sykophantēs</i>) ‘voluntary denouncer’; ἀκρόπολις (<i>akropolis</i>) ‘upper city’; πόλεμος (<i>polemos</i>) ‘war’; Ἀθήνη (<i>Athēnē</i>) ‘Athens’; Περικλῆς (<i>Periklēs</i>) ‘Pericles’; Σωκράτης (<i>Sōkratēs</i>) ‘Socrates’; Παναθήναια (<i>Panathēnaia</i>) ‘Panathenaea’; ἐκκλησία (<i>ekklēsia</i>) ‘assembly’

namely count-based models, word embeddings and syntactic word embeddings. We explained how words can be located in a semantic space from vector representations computed by these models. It then becomes possible to measure the semantic distance between words by measuring the distance between their vectors. Using vector representations to study word semantics and semantic change has the advantage of being able to process large quantities of data in a short time, efficiently computing complex calculations. The results obtained via language modelling are indeed co-occurrence based, but the models store more complex information than just retrieving concordances. We showed examples of semantic information that can be extracted from language models, i.e., the most strictly related words in the semantic space to a given target word (nearest neighbours) and a coefficient quantifying the level of semantic relatedness between pairs of words (cosine similarity). The performance of the specific models we adopted needs to be assessed against a gold standard in order to have an idea of their reliability. We have shown an example of an evaluation of different models against a benchmark that has been developed as a gold standard for Ancient Greek, AGREE. We also showed preliminary results obtained by incorporating syntactic information during the training of word embeddings.

We also mentioned some limitations of this approach. First of all, language modelling requires a large quantity of training data, which is often lacking for historical languages. Second, since they work on a statistical basis, language models do not provide reliable results for low-frequency words. Moreover, different architectures might be necessary for different tasks since they do not capture all

the same kind of information and the same benchmark may not be adequate to evaluate the performance of all kinds of models. This was shown by an evaluation of the word embeddings incorporating syntactic information against AGREE, developed to evaluate semantic relatedness. The problem here is that syntactic word embeddings capture functional rather than topical similarity, which is not reflected in the AGREE benchmark. Finally, for the specific purpose of detecting and studying semantic change, a critical approach towards the detected changes is needed since some of the observed changes could be due to the composition of the diachronic corpus rather than being real modifications in word meaning through time.

Funding

This work was partially supported by the Young Academy Groningen through Silvia Stopponi's PhD scholarship. This article was made possible through the financial support of Anchoring Innovation. Anchoring innovation is the Gravitation Grant research agenda of the Dutch National Research School in Classical Studies, OIKOS. It is financially supported by the Dutch Ministry of Education, Culture and Science (NWO project number 024.003.012). For more information about the research program and its results, see the website www.anchoringinnovation.nl.

Open Access publication of this article was funded through a Transformative Agreement with University of Groningen.

Acknowledgements

We thank the two anonymous reviewers whose comments helped us to improve this article.

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Abbreviations

ADV	adverbial phrase
AGREE	Ancient Greek Relatedness Embeddings Evaluation
CADE	Compass-aligned Distributional Embeddings
CBOW	Continuous Bag-of-Words
GASC	Genre-Aware Semantic Change
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
NP	noun phrase
PPMI	Positive Pointwise Mutual Information
PROIEL	Pragmatic Resources in Old Indo-European Languages
TLG	Thesaurus Linguae Graecae
t-SNE	t-distributed Stochastic Neighbor Embedding
VP	verbal phrase
XADV	adverbial phrase sharing an argument with the main clause

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Résumé

Les méthodes computationnelles ont produit des résultats significatifs et utilisables pour étudier la sémantique des mots, y compris le changement sémantique. Ces méthodes, qui appartiennent au domaine du traitement automatique des langues, ont été appliquées récemment aux langues anciennes. Notamment, la modélisation du langage a été appliquée au grec ancien, la langue sur laquelle nous nous concentrons. Dans cette contribution on explique comment des vecteurs de mots peuvent être calculés à partir de cooccurrences dans un corpus et comment ils peuvent être utilisés pour localiser les mots dans un espace sémantique. On explique aussi quel type d'information peut être extrait des modèles de langage. On compare trois différents types de modèles de langue qui peuvent être utilisés pour étudier la sémantique du grec ancien: un modèle à décomptage (count-based), un modèle Word2vec (qui produit des

plongements de mots, ‘word embeddings’) et des plongements de mots enrichis d’information syntactique. On présente des exemples montrant comment la qualité de ces représentations de mots peut être évaluée. On met en évidence les avantages et les potentialités de ces méthodes, notamment pour étudier le changement sémantique, ainsi que leurs limites.

Zusammenfassung

Es hat sich gezeigt, dass rechnerische Methoden aussagekräftige und nutzbare Ergebnisse für die Untersuchung der Wortsemantik, einschließlich semantischer Veränderungen, liefern. Diese Methoden, die zum Bereich des Natural Language Processing gehören, werden seit kurzem auf alte Sprachen angewendet, Insbesondere auf das Altgriechische, das hier von Interesse ist. In diesem Beitrag erklären wir, wie Vektordarstellungen aus Wort-Kookkurrenzen in einem Korpus berechnet und zur Lokalisierung von Wörtern in einem semantischen Raum verwendet werden können und welche Art semantischer Informationen aus Sprachmodellen extrahiert werden können. Wir vergleichen drei verschiedene Arten von Sprachmodellen, die zur Untersuchung altgriechischer Semantik angewendet werden können: ein zählbasiertes Modell, ein Worteinbettungsmodell und ein syntaktisches Einbettungsmodell, und wir zeigen an Beispielen, wie die Qualität ihrer Darstellungen bewertet werden kann. Wir zeigen die Vorteile und das Potenzial dieser Methoden auf, insbesondere für die Untersuchung semantischer Veränderungen, beleuchten aber auch deren Grenzen.

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Publication history

Date received: 27 February 2023
Date accepted: 14 October 2023
Published online: 2 July 2024